HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming

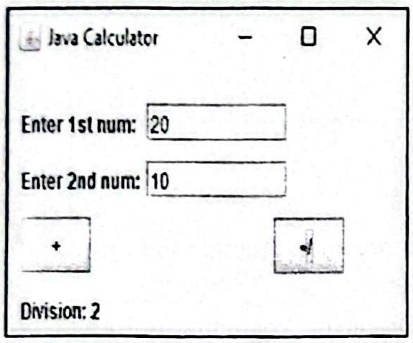
Batch: 27th | Semester: Spring 2026
Section: I | Time: 2 Hours
Date: 03/09/2026 | **Final Examination** | Total Marks: 40

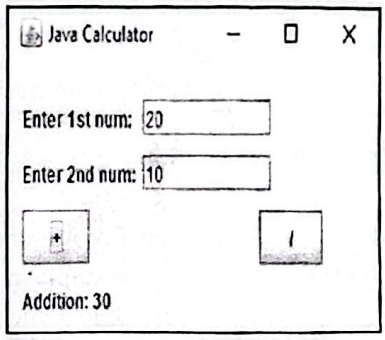
Answer any five questions from the following. $5 \times 8 = 40$

			CLOs	Marks
1	(a).	What is a 'thread' in Java? How does the multithreading upgrade the CPU utilization? Explain with an example in brief.	CLO3	3
	(b).	Explain the life cycle of a thread in Object Oriented Programming (OOP).	CLO2	5
2	(a).	Consider the string "**MadAm**". Write a program in java to find if the given string is palindrome or not. Consider, the string is case insensitive.	CLO3	4
	(b).	Write a Java program to get the character at the given index within the string. Sample input/output: ***Original String = Java Exercises!*** ***The character at position 0 is J*** ***The character at position 10 is i***	CLO3	4
3	(a).	What is an Exception in OOP? Explain with an example.	CLO1	3
	(b).	Write a program in Java for '**ArithmeticException**' and '**ArrayIndexOutOfBoundsException**'.	CLO2	5
4	(a).	Write a program in Java to access constructor of an abstract class.	CLO1	3
	(b).	Write a Java program to create an abstract class Animal with an abstract method called **Sound()**. Create subclasses **Lion** and **Tiger** that extend the Animal class and implement the **Sound()** method to make a specific sound for each animal.	CLO2	5
5	(a).	Briefly explain Java Interface with an example.	CLO3	3
	(b).	Write a Java program to create an interface **Shape** with the **getArea()** method. Create three classes Rectangle, Circle, and Triangle that implement the Shape interface. Implement the **getArea()** method for each of the three classes.	CLO3	5
6	(a).	Write four **ActionListener** built-in method names.	CLO4	2

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	(b).	Define **Abstraction** and **Encapsulation** in Object Oriented Programming.	CLO1	2
	(c).	Explain the **StringBuffer** class in Java.	CLO3	4
7	(a).	What is the use of an *import* declaration in a Java program?	CLO5	2
	(b).	Write a Java program to display the following interface using **AWT, Swing** class with **ActionListener**.	CLO4	6





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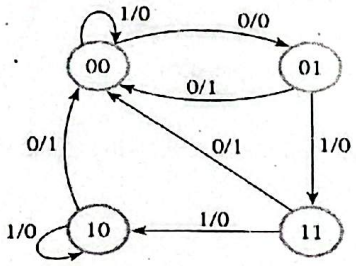
HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE225: Digital Logic Design

Batch: 27th | Semester: Spring 2026
Date: 06/09/2026 | **Final Examination** | Time: 2 Hours
Total Marks: 40

Answer any **five** questions of the following. $5 \times 8 = 40$

1.	(a)	Define a combinational logic circuit and explain how it differs from a sequential logic circuit.	CLO1	3
	(b)	Design a Full Adder using two Half Adders and an OR gate. Construct its truth table and derive the Boolean expressions for the Sum and Carry outputs using a K-map.	CLO2 & CLO4	5
2.	(a)	Design a 1-bit magnitude comparator showing the Boolean expressions for A>B, A=B, and A<B with truth table.	CLO4	3
	(b)	What is the toggling problem in a latch? Explain why it can make the output unpredictable.	CLO1	2
	(c)	Construct an SR latches using NAND and NOR gates, and verify the truth tables for all possible input combinations.	CLO1	3
3.	(a)	Design a JK flip-flop and derive its characteristic table and excitation table from its truth table.	CLO1	4
	(b)	Demonstrate how a SR flip-flop can be converted into a JK flip-flop.	CLO1	4
4.	(a)	For the given state diagram, derive the equations from the excitation table required to design a clocked sequential circuit using a D flip-flop.	CLO1	4

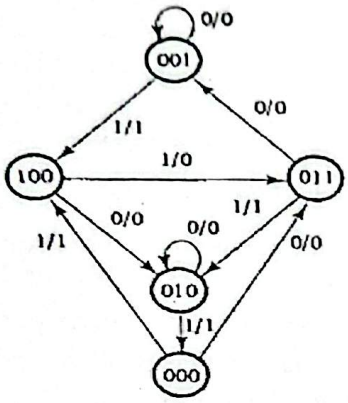


	(b)	Reduce the number of states in the following state table and construct the corresponding reduced state table.	CLO1	4

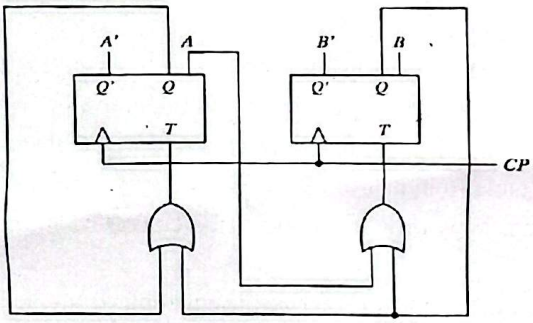
Present State	Next State $x=0$	Next State $x=1$	Output $x=0$	Output $x=1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

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5. (a) Perform state assignment and derive the state table for the following state diagram. CLO1 3



(b) For the given sequential circuit, analyze the possible state transitions and output sequences and finally draw a state diagram. CLO1 5



6. (a) Explain the differences between Moore and Mealy state machines. CLO1 2

(b) Design a 3-bit asynchronous DOWN counter with JK flip-flops and provide a logical explanation of its operation. CLO1 3

(c) Design a 3-bit UP/DOWN ripple counter with necessary circuit derivation. CLO1 3

7. (a) Design a PAL for implementing the following equations. CLO5 3

$A = \sum m(8, 9, 12, 13)$

$B = \sum m(4, 5, 12, 13)$

$C = \sum m(5, 7, 13, 15)$

(b) Given the Boolean functions $Y_1=AB'$ and $Y_2=(A+B)\,C'$, design a PLA that implements these functions. CLO5 2

(c) Design the block diagram of a 16×8 ROM and explain its working. CLO5 3

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HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: ENS 221 | **Course Title: Environmental Science**
Final Examination | Semester: Spring 2026
Date: 9/10/2026 | Total Time: 02:00 Hours | **Total Marks: 40**

*** If necessary, draw relevant graphs, diagrams to support your explanations.:** $4 \times 10 = 40$
[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

			CLOs	Marks
		Part A (Answer any Two)		
Q1.	**(a)**	Define water pollution and identify two major sources of water pollution.	CLO1	[1]
	(b)	Explain environmental pollutants and distinguish between natural and anthropogenic pollutants with relevant examples.	CLO2	[4]
	(c)	Analyze water-pollution pressures in Bangladesh from industrial discharge, sewage, plastics and agricultural runoff. Propose technology-supported measures for monitoring and control.	CLO4	[5]
Q2.	**(a)**	What is green tourism? Mention two characteristics of environmentally responsible tourism.	CLO1	[1]
	(b)	Explain how ecotourism can affect local culture, biodiversity and natural resources.	CLO2	[4]
	(c)	Evaluate barriers to sustainable ecotourism in Bangladesh and propose technology-based and community-based solutions for a destination such as the Sundarbans or Cox's Bazar.	CLO4	[5]
Q3.	**(a)**	What are the SDGs?	CLO1	[1]
	(b)	Explain the relevance of SDG 6, SDG 9, SDG 12 and SDG 13 to computer science, digital innovation and technological development.	CLO2	[4]
	(c)	How can data, AI, IoT and digital platforms support sustainable development while reducing environmental impacts?	CLO3	[5]
		Part B (Answer any Two)		
Q4.	**(a)**	Define disaster mitigation and explain its purpose.	CLO1	[2]
	(b)	Assess the effectiveness of cyclone preparedness and EWS in Bangladesh.	CLO2	[3]
	(c)	How can mobile networks, remote sensing, IoT sensors, satellite data and digital communication improve preparedness?	CLO3	[5]
Q5.	**(a)**	What is sea-level rise (SLR)?	CLO1	[1]
	(b)	Analyze how climate change can intensify water stress, floods, heatwaves and coastal risks in Asia.	CLO2	[3]

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	(c)	Explain how Bangladesh can use climate information, GIS/remote sensing and data-driven early-warning systems to improve resilience.	CLO2	[6]
Q6.	(a)	What is the role of an early-warning system (EWS) in disaster management?	CLO1	[2]
	(b)	Describe the Six Pillars of the BCCSAP.	CLO3	[3]
	(c)	Discuss how digital technologies, data systems and local institutions can support climate adaptation and disaster-risk reduction	CLO4	[5]

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HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 229: Computer Architecture

Batch: 25th
Date: 08.09.26

Final-Term Examination

Time: 2 Hours
Total Marks: 40

Answer any five questions. 5 × 8 = 40

1.	(a)	Distinguish between Computer Architecture and Computer Organization with example.	CLO1	2
	(b)	Draw the internal organization of a 2M * 8 dynamic memory chip.	CLO1	6
2.	(a)	Define RAM. Mention the different types of RAM.	CLO4	2
	(b)	Draw the diagram of synchronous DRAM	CLO4	6
3.		Write about EEPROM.		
	(a)		CLO4	2
	(b)	Write down the advantages and disadvantages of the SJF algorithm.	CLO4	6
4.	(a)	Draw the diagram of Pentium 4 block .	CLO4	4
	(b)	Explain three replacement algorithms.	CLO4	4
5.	(a)	What is a hazard?	CLO3	2
	(b)	Write down the types of hazards. To overcome these hazard which steps would you take?	CLO3	6
6	(a)	Draw the flowchart of a Six Stage Instruction Pipeline .	CLO3	4
	(b)	Define Instruction pipeline. Write down the steps of the Instruction pipeline.	CLO3	4
7.	(a)	Write about the Membrane Key switch and Hall effect Key switch with proper diagram.	CLO3	2
	(b)	Write down the differences between impact printers and non impact printers.	CLO4	2
	(c)	Write down the advantages and disadvantages of dot matrix printers.	CLO3	4

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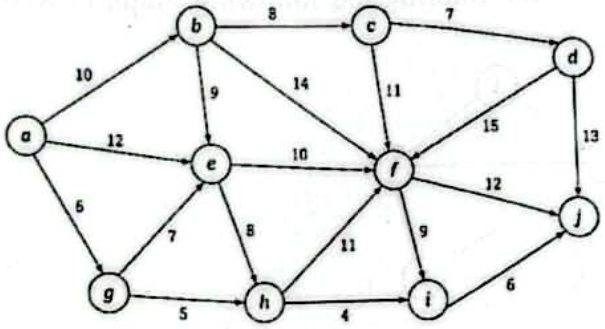
Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE221: Computer Algorithms
Final Examination
Semester: Spring 2026

Batch: 27th
Date: 01-09-2026

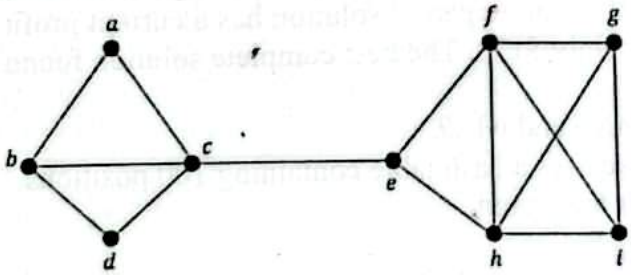
Time: 2 Hours
Total Marks: 40
$5 \times 8 = 40$

Answer any **five** questions of the followings.

			CLOs	Marks
1.	(a)	Write the differences between Greedy Method and Dynamic Programming.	CLO4	3
	(b)	Apply Dijkstra's Algorithm to find the shortest path from the node 'a' to all other nodes in the graph.	CLO3	5



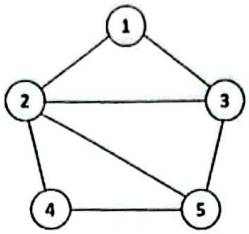
2.	(a)	A social media platform wants to find all connected users starting from one user. Explain how a Breadth-First Search (BFS) can help.	CLO2	2
	(b)	Find the articulation points of the following graph using a Depth-First Search (DFS) spanning tree.	CLO3	6



3.	(a)	Suppose, a warehouse manager wants to maximize profit by selecting products to load into a truck with a limited weight capacity. Each product has a weight and profit value. He must decide which products to include. Which Dynamic Programming approach should be used, and why?	CLO2	3
	(b)	Consider the following instance of the 0-1 knapsack problem:	CLO3	5

Number of items, $n = 4$,
Capacity, $W = 11$,
Profits $\{p_1, p_2, p_3, p_4\} = \{7, 18, 11, 9\}$ *and*
Weights $\{w_1, w_2, w_3, w_4\} = \{3, 4, 7, 4\}$
Calculate the optimal solution using dynamic programming.

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4.	(a)	A software engineer is developing a program to compute the nth Fibonacci number. The recursive version becomes very slow for a large value of n. Compare Recursion and Dynamic Programming for this scenario.	CLO4	3
	(b)	Consider the strings X= "HARPTULXRD" and Y= "HYRPGULZED". If each insertion and deletion have a unit cost and a replace cost 2 units, then generate the cost table to calculate the minimum distance to convert X into Y.	CLO3	5
5.	(a)	A programmer is placing queens on a chessboard. After placing several queens, the next queen cannot be placed in any safe position. What should the algorithm do, and why?	CLO4	2
	(b)	Find out three possible solutions for coloring the following graph in RGB color mode: 	CLO3	6
6.	(a)	Explain the basic principle of Branch and Bound technique.	CLO2	2
	(b)	A delivery company has several jobs, each with a deadline and profit. The company uses Branch and Bound to find the maximum-profit schedule. During the search, one partial solution has a current profit of $180, while its calculated upper bound is $220. Another partial solution has a current profit of $150, but its upper bound is only $170. The best complete solution found so far has a profit of $190. Which branch should be discarded and why?	CLO3	3
	(c)	A company's employee database uses a hash table containing 100 positions. Currently, 80 employee records are stored. i. Calculate the load factor. ii. What problem may occur because of this load factor? iii. Suggest a suitable solution.	CLO2	3
7.	(a)	What is Lower Bound Theory?	CLO2	2
	(b)	What is a decision tree? How is it used to determine a lower bound?	CLO2	3
	(c)	Prove that, any decision tree that can sort n elements must have $\Omega(nlogn)$ height.	CLO3	3

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Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 27th | Semester: Spring 2026
Date: 13/09/2026 | Final Examination | Time: 2 Hours
Total Marks : 40

Answer any **five** questions: $5 \times 8 = 40$

		CLOs	Marks
1.	(a) Illustrate the components of pulse code modulation (PCM) encoder.	CLO5	4
	(b) Explain two common scrambling techniques.	CLO2	4
2.	(a) Draw and describe amplitude modulation in analog-to-analog conversion.	CLO5	4
	(b) Briefly explain data transmission and its modes.	CLO4	4
3.	(a) Define constellation diagram and explain its role in analog transmission.	CLO5	4
	(b) Explain Quadrature Amplitude Modulation.	CLO5	4
4.	(a) List all categories of multiplexing. Define FDM.	CLO5	3
	(b) Illustrate three strategies of time division multiplexing.	CLO5	5
5.	(a) Describe spread spectrum and its goal.	CLO5	4
	(b) Draw and explain digital signal service (Digital hierarchy).	CLO4	4
6.	(a) How does sky propagation differ from line-of-sight propagation? Explain.	CLO1	4
	(b) Write the differences between **omnidirectional** waves and **unidirectional** waves.	CLO1	4
7.	(a) Explain packet switching network.	CLO4	
	(b) Describe three phases of circuit-switched network.	CLO4	

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